

GTFShift: A Data-Driven Tool for Prioritizing Bus Lane Implementation

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Abstract

Identifying the streets with the highest density of bus frequency can help in planning and prioritizing the implementation of bus lanes. GTFShift R package uses the General Transit Feed Specification (GTFS) to perform this analysis, allowing for an aggregated overview of the transit density over a given territory.

Keywords: Bus lanes, GTFS, Network prioritization, R package.

1 Questions

Bus lanes (or Transit priority lanes) have the potential to improve bus operation reliability by limiting the negative impacts of traffic congestion (Cesme et al., 2018), reducing the variability of its running time (Surprenant-Legault & El-Geneidy, 2011), increasing the average commercial speed that can ultimately be used to increase the frequency of service without additional material or human resources (Basso et al., 2011). Benefits multiplied by a contiguous network, with the highest savings in travel time verified at intersections (Truong et al., 2015).

However, introducing them on heterogeneous traffic infrastructure has an impact on other modes, such as private vehicles, that with less space allocated, witness a decrease in the level of service (Arasan & Vedagiri, 2010), jeopardizing public acceptance (Batty et al., 2014).

The rules of thumb for implementing bus lanes are not well defined, but some common criteria include:

- Hourly frequency of buses and trams at a peak hour
- Number of lanes in the same direction
- Existing bus lanes in the area (from a network perspective)

For instance, NACTO suggests that dedicated bus lanes should be applied «on major routes with frequent headways (10 minutes at peak) or where traffic congestion may significantly affect reliability» (NACTO, 2016). Nevertheless, the decision should be based on a more detailed analysis of the network, including the existing traffic conditions, the number of bus stops, and the overall demand for public transport in the area. Vuchic (1981) defends that bus lanes should be implemented based on the number of lanes, vehicle volume per lane, and bus frequency per hour. For instance, the minimum bus frequency for a street with 2 lanes in the same direction and 400 veh/h/lane is considered to be 12 bus/h to justify its implementation.

Identifying the corridors with highest bus frequency in the road network is then critical to prioritize new bus lanes and justify its implementation.

To support this analysis, this paper presents **GTFShift**, an open-source R package to identify priorities for bus lanes' implementation (Matos & Félix, 2025). Based on the analysis of the General Transit Feed Specification (GTFS) data, it extends the capabilities of other R libraries that deal with this transit standard, such as **tidytransit** (Poletti et al., 2025) and **gtfstools** (Herszenhut et al., 2025).

This tool can be used to address the prioritization of bus lanes for a given territory in order to increase its transit efficiency.

2 Methods

GTFSshift is an open-source R package that provides methods for the entire workflow of bus network frequency analysis. It can be used to get GTFS data for a given territory, and estimate the aggregated bus frequency per street segment at each hour of a given day.

- `unify()`: Merges multiple GTFS feeds into a single entity, enabling an aggregated analysis. This method is relevant, for instance, when considering more than one bus operator in the same territory. It extends the functionality of `gtfstools::merge_gtfs()` by enabling the identification of additional transfers.
- `get_route_frequency_hourly()`: Calculates the hourly frequency of bus routes for a given day, considering its geometries. By default, it estimates the next business Wednesday, relevant for the peak hour. Using `overline` parameter aggregates all the bus routes that share the same geometry, providing a single frequency value per street segment, using `stplanr::overline2()` (Lovelace & Ellison, 2018). Optionally, using `use_osm_routes` parameter, it retrieves the geometries from *OpenStreetMap* (by matching the tag `gtfs:shape_id`), overwriting the original GTFS `shapes.txt`.

The parameter `use_osm_routes` is particularly useful if the GTFS shapes do not share the same geometry. For instance, if the edges of the lines do not overlap or do not follow the same route-over-the-road – which is very common, even besides [GTFS recommendation](#) – geometries might not be aggregated correctly, causing inconsistent results. By relying on a common road network, such as OSM, it is possible to overcome this issue and aggregate the bus routes correctly. Nevertheless, the `gtfs:shape_id` OSM tag is only present in 3.5% of the mapped bus networks. Alternatively, the `osm_shapes_match_routes()` method can be used to match GTFS shapes with OSM networks, considering their geometrical resemblance.

The case study is Lisbon, Portugal, where the bus network is operated by *Carris* and *Carris Metropolitana*, respectively, the urban and sub-urban bus operators in the area.

For this article, we used GTFSshift version v0.6 and R version v4.5. To reproduce the results presented in this paper, refer to [U-Shift/busclar](#) code repository.

3 Findings

We start by loading the R libraries, the static GTFS feeds for both bus operators and merging them.

```
library(GTFSshift)

# load GTFS feeds
gtfs_carris = load_feed("https://gateway.carris.pt/gateway/gtfs/api/v2.8/GTFS") # 114 MB
gtfs_carrismetropolitana = load_feed("https://api.carrismetropolitana.pt/gtfs") # 638 MB
```

```
# unify the feeds in a single GTFS feed
gtfs_lisbon = unify(gtfs_carris, gtfs_carrismetropolitana, create_transfers = FALSE)
```

Next, the bus routes frequency is estimated hourly for the next business Wednesday, and the geometries are overlined, considering the correspondent OSM routes.

```
# OSM query for bus networks in Lisbon
library(osmdata)
q = opq("Lisbon") |>
  add_osm_feature(key = "route", value = "bus") |>
  add_osm_feature(key = "network", value = c("Carris", "Carris Metropolitana")) # operators

# get hourly frequency by shape
frequencies_segment = get_route_frequency_hourly(gtfs_lisbon,
```

```
overline = TRUE,
use_osm_routes = q)
```

The results, presented in Figure 1, allow for the identification of the street segments with higher bus transit frequency, at morning peak hour.

```
# filter for the peak hour 8-9am
frequencies_segment_8 = frequencies_segment |>
  dplyr::filter(hour == 8)

mapview::mapview(
  frequencies_segment_8,
  zcol = "frequency",
  lwd = "frequency",
  layer.name = "Bus frequency"
)
```

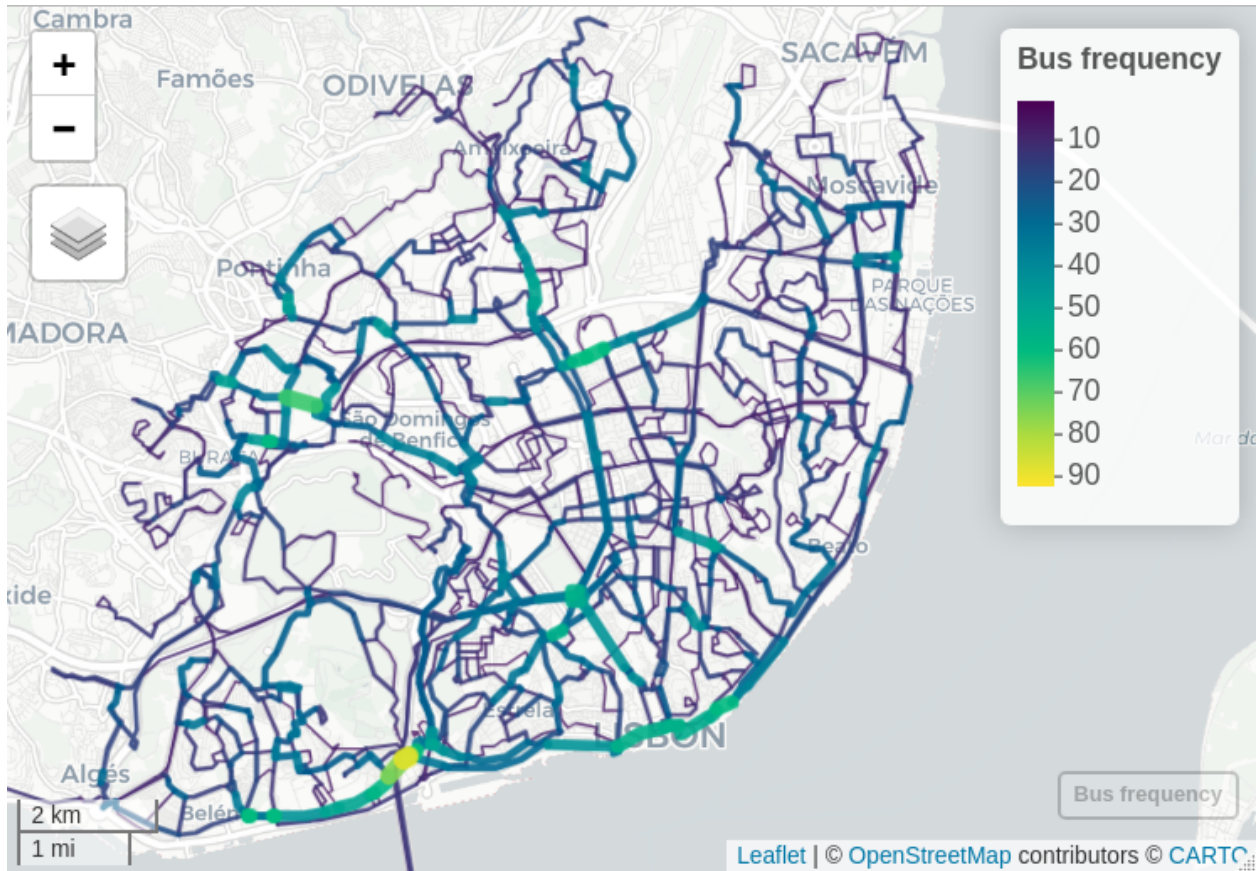


Fig. 1. Road segments with higher bus frequency, in Lisbon, at 8 am. The larger and yellowish lines represent the streets with higher hourly bus frequency. Carto basemap.

The presented replicable method provides an feasible methodology to generate evidence to support decision-making when considering the bus lane network expansion, for a given city. The data requirements are commonly available for most cities, as GTFS feeds are widely used by public transport operators. We acknowledge that this evidence may not be sufficient to justify the implementation of bus lanes, as it does not consider other factors such as traffic conditions, public support, and roadway capacity.

Additionally, filtering the road network segments by existing number of lanes in the same direction (2 or more) may better support the decision of implementing bus corridors. Public policies may opt to implement bus lanes in streets with only one lane, or implement dynamic bus lanes (for peak hours only), for instance. This method also allows to consider the implementation of dynamic bus lanes (instead of static), by taking into account the hourly frequency – providing evidence for decision on which hours to allocate road lanes for bus.

Further enhancements of the method may include the passenger capacity per hour, instead of the frequency of the number of vehicles. For instance, bus routes with articulated buses (more than 80 passengers) would weight more than an historical tram route (max 25 passengers).

Bus lanes for emergency vehicles, in the approach to hospitals, for instance, could also be considered.

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References

- Arasan, V. T., & Vedagiri, P. (2010, 5). Study of the impact of exclusive bus lane under highly heterogeneous traffic condition. *Public Transport*, 2(1-2), 135–155. doi: 10.1007/s12469-010-0021-x
- Basso, L. J., Guevara, C. A., Gschwender, A., & Fuster, M. (2011, 2). Congestion pricing, transit subsidies and dedicated bus lanes: Efficient and practical solutions to congestion. *Transport Policy*, 18(5), 676–684. doi: 10.1016/j.tranpol.2011.01.002
- Batty, P., Palacin, R., & González-Gil, A. (2014, 12). Challenges and opportunities in developing urban modal shift. *Travel Behaviour and Society*, 2(2), 109–123. doi: 10.1016/j.tbs.2014.12.001
- Cesme, B., Roisman, R., Burns, R., List, K., Koudounas, A., Cuellar, J., ... Miller, D. (2018, 12). Strategies and barriers in effective bus lane implementation and management: Best practices for use in the greater washington, d.c. region. *Transportation Research Record: Journal of the Transportation Research Board*, 2672(8), 29–40. doi: 10.1177/0361198118791914
- Herszenhut, D., Pereira, R. H. M., Andrade, P. R., & Bazzo, J. (2025). gtfstools: General transit feed specification (GTFS) editing and analysing tools [Computer software manual]. Retrieved from <https://ipeagit.github.io/gtfstools/> (R package version 1.4.0, <https://github.com/ipeaGIT/gtfstools>)
- Lovelace, R., & Ellison, R. (2018). stplanr: A package for transport planning. *The R Journal*, 10(2), 10. doi: 10.32614/RJ-2018-053
- Matos, G., & Félix, R. (2025). GTFShift: Explore and analyse general transit feed specification (GTFS) files with a focus on urban mobility [Computer software manual]. Retrieved from <https://github.com/U-Shift/GTFShift> (R package version 0.6.0)
- NACTO. (2016). *Transit Street Design Guide*. Island Press.
- Poletti, F., Herszenhut, D., Padgham, M., Buckley, T., & Noriega-Goodwin, D. (2025). tidytransit: Read, validate, analyze, and map gtfs feeds [Computer software manual]. Retrieved from <https://github.com/r-transit/tidytransit> (R package version 1.7.0)
- Surprenant-Legault, J., & El-Geneidy, A. M. (2011, 1). Introduction of reserved bus lane. *Transportation Research Record Journal of the Transportation Research Board*, 2218(1), 10–18. doi: 10.3141/2218-02
- Truong, L. T., Sarvi, M., & Currie, G. (2015, 1). Exploring multiplier effects generated by bus lane combinations. *Transportation Research Record Journal of the Transportation Research Board*, 2533(1), 68–77. doi: 10.3141/2533-08

Vuchic, V. R. (1981). *Urban Public Transportation - Systems and Technology*. Prentice-Hall, Inc.